

TrES-3b

R-2740

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140609 · created 2026-09-02T01:18:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456817.8775 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.015
Transit depth (Rp/Rs)^2	3.0 +/- 0.53 [%]
Orbital Inclination (inc)	82.66 +/- 0.77
Airmass coefficient 1 (a1)	1.345 +/- 0.014
Airmass coefficient 2 (a2)	-0.284 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	None
Optimal Aperture	3.6
Optimal Annulus	14.31
Transit Duration (day)	0.062 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.173 $\pm$ 0.015 vs 0.16309 (deviation 0.66 σ)
Duration fit vs published	0.062 d vs 0.0567 d (deviation 0.87 σ)
Mid-time O–C	0.7 min
Depth SNR	11.5
Residual scatter	1.02 %

Light curve

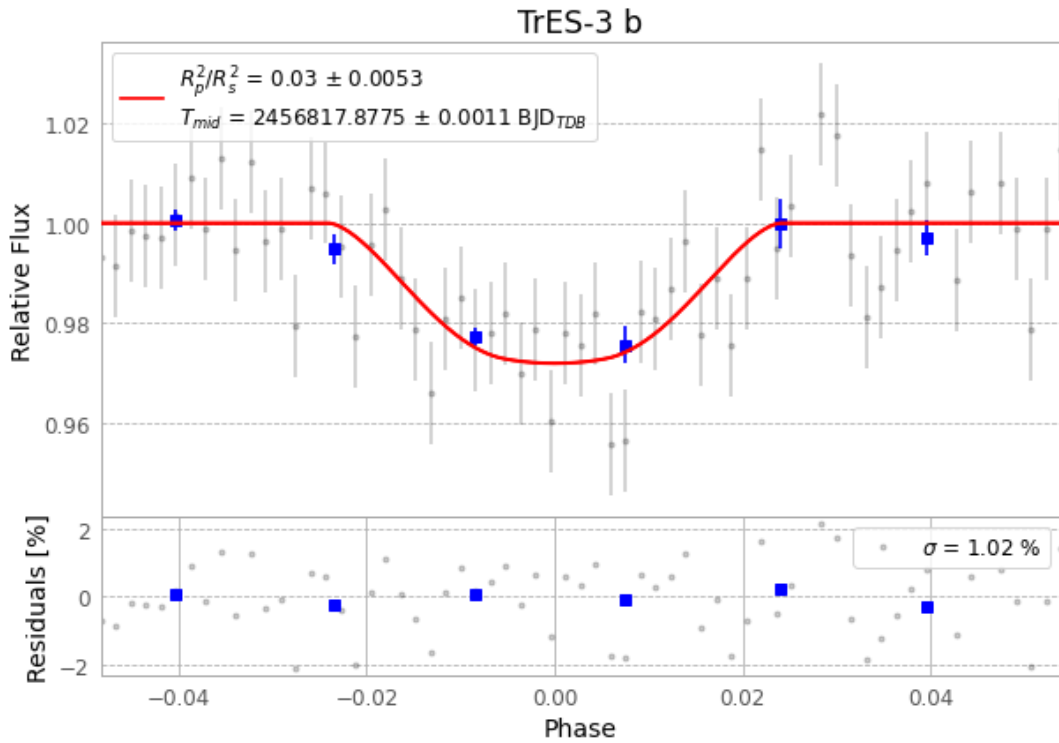


Figure 1 — FinalLightCurve_TrES-3 b_2014-06-09.png (SHA-256 dd324297f62b9443...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 283], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8009f5cb1e58fe1f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3140609072713.FITS → TRES-3140609103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140609.log (531e16971e19...), FinalLightCurve_TrES-3 b_2014-06-09.png (dd324297f62b...), FinalLightCurve_TrES-3 b_2014-06-09.csv (fe427db35824...)

Compute resources

Wall time 151.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:18Z.